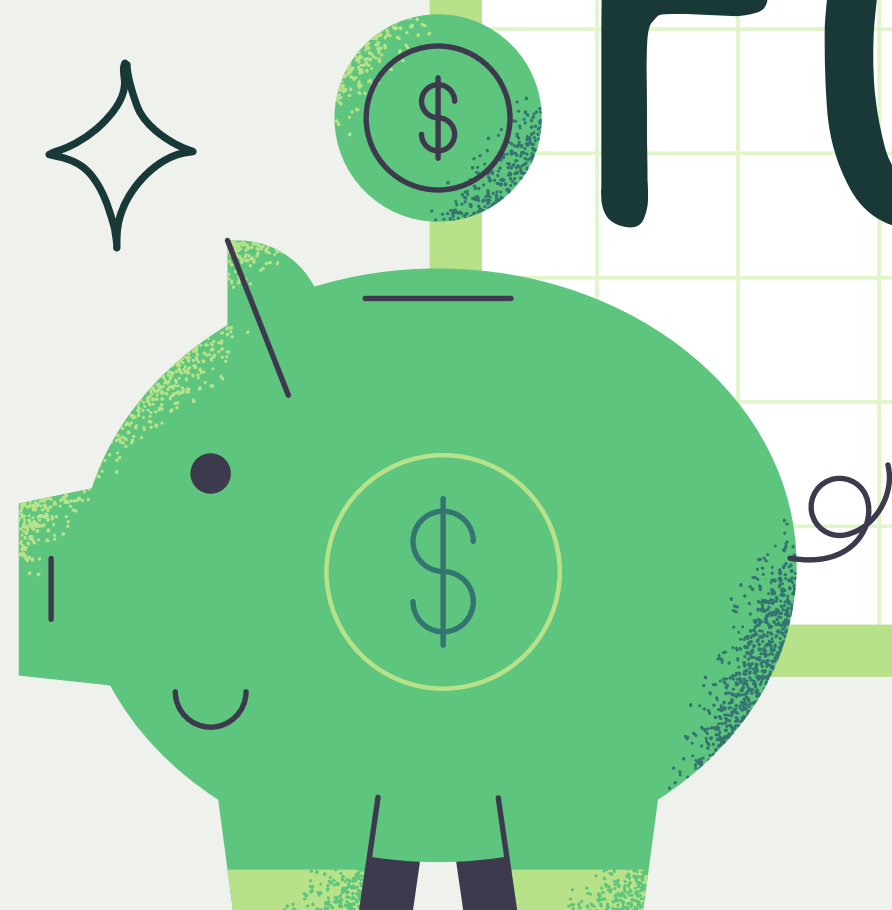
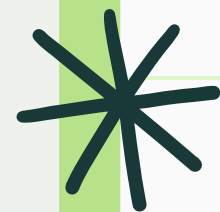
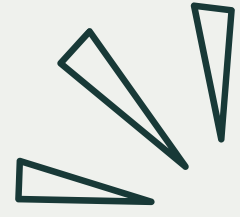




INTRODUCTION

Time series forecasting is widely used in domains such as finance, energy systems, and weather prediction. This project introduces practical techniques for analyzing temporal data and building forecasting models using machine learning and deep learning. Students will also explore a multi-model forecasting framework, where different models act as strategic agents whose predictions are combined using game-theoretic aggregation methods to produce robust forecasts.





WHY THIS PROJECT?

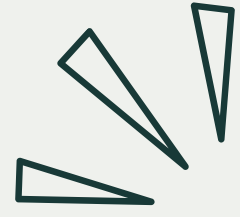


- **STAGE 1: Foundations of Time Series Analysis**

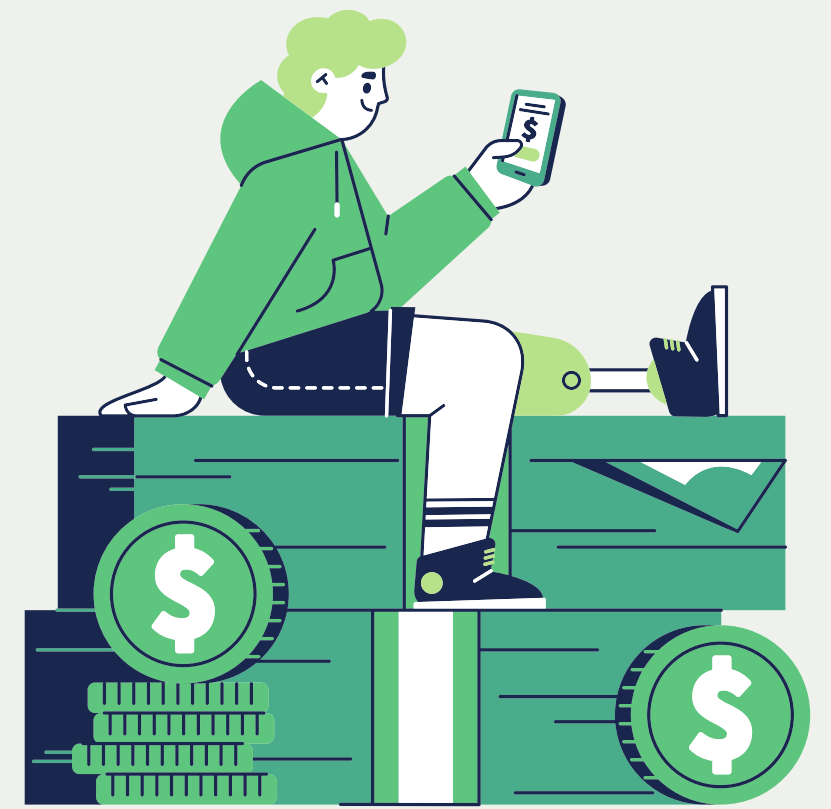
- Understand key components of time series: trend, seasonality, and noise.
- Visualize and explore temporal datasets using NumPy, Pandas, and Matplotlib.
- Apply smoothing techniques such as moving averages.
- Create useful features like lag features and rolling statistics.

- **STAGE 2: Forecasting with Machine Learning and Deep Learning**

- Train basic forecasting models using machine learning algorithms.
- Implement deep learning models (LSTM/GRU) for sequence prediction.
- Evaluate and compare model performance using forecasting metrics.
- Understand strengths and limitations of different models.



WHY THIS PROJECT?



- **Stage 3: Strategic Multi-Model Forecasting**

- Understand the concept of ensemble forecasting and model aggregation.
- Interpret forecasting models as agents competing for predictive accuracy.
- Implement a game-theoretic weighting mechanism to combine predictions from multiple models.
- Build a complete multi-agent forecasting pipeline from data preprocessing to final prediction.
- Evaluate how strategic model combination improves forecasting robustness.



TIMELINE



- **WEEK 1**

Introduction to time series data and basic analysis using NumPy, Pandas, and Matplotlib.

- **WEEK 2**

Understanding time series components such as trend, seasonality, and smoothing using moving averages.

- **WEEK 3**

Feature engineering for time series including lag features, rolling statistics, and temporal features.

- **WEEK 4**

Building baseline forecasting models and launching the Kaggle-style competition dataset and evaluation metric.



TIMELINE



- **WEEK 5**

Applying machine learning models such as Linear Regression and Random Forest to improve competition performance.

- **WEEK 6**

Introduction to deep learning for sequence modeling using LSTM/GRU networks for forecasting tasks.

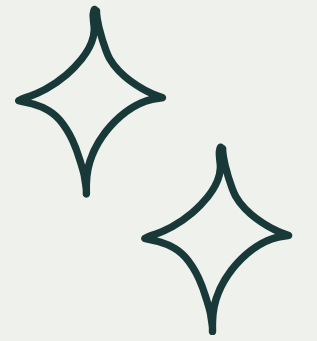
- **WEEK 7**

Ensemble forecasting and introduction to game-theoretic ideas for combining multiple model predictions.

- **WEEK 8**

submission and evaluation of the Kaggle competition leaderboard along with discussion of the best forecasting strategies.

MENTORS' CONTACT



Mentor 1

Manthan Khetade

manthanvk24@iitk.ac.in
8766477558



Mentor 2

Aayushman Tripathi

aayushmant24@iitk.ac.in
6395629200



Mentor 3

Arisht Daiya

arishtd24@iitk.ac.in
9660267131



- No strict prerequisites – just basic Python knowledge is recommended. Familiarity with NumPy and Pandas would be helpful, but you can pick it up along the way.





THANK YOU

